

Symbolic mode uses mathematical symbols to perform the changes in permissions, these are the following mathematical symbols used for following permission operations:-

- + is used for adding permissions for a specific user
- - is used for removing permissions for a specific user
- = is used for overriding existing permissions with new values

Syntax for changing File and Directory permissions in symbolic mode:-

`Chmod [class][operator][permission] file`

Eg: `chmod [ugoa][+ or -][rwx] chris.txt`

Now let's take a look at some examples to get the concept better

Consider a file `chris.txt` with file permissions `-rw-rw-r--`, you have to change these permissions so that owner has full permissions, group has read and others have no permissions. We will do the required task in three steps.

First we will add execution permission for the user:

`chmod u+x Chris.txt`

Now we will remove the write permissions from the group:

`chmod g-w Chris.txt`

Lastly we will remove the read permission from others:

`chmod o-r Chris.txt`

This was just to make you understand the step by step process, you can even do this whole task in one go with the following command:

`chmod u+x,g-w,O-r Chris.txt`

Note: You should always remember that each section is separated by commas and there is no gap in between.

This might seem all too difficult to implement with all these numerals and symbols involved over these two modes of `chmod`, you don't need to worry about that at all, just get familiarized with the topic and you will remember them.

Changing File ownership in Ubuntu

For changing the ownership of a file or a directory the common `chown` command is used. You might have guessed it by now that `chown` stands for change owner, which is actually the correct full form.

Every file has a specific user ID or UID and a specific group ID or GID, the `chown` command is used to change the user owner as well as the user and group owners of a file.

Basic Syntax for using a `chown` command:

`chown user:group filename`